

Multiple Choice (10 marks)

1. What is the greatest common factor of 36 and 90?
 - (a) 6
 - (b) 18
 - (c) 36
 - (d) 180
2. Given that the sample has a standard deviation of zero, which of the following is a true statement?
 - (a) The standard deviation of the population is also zero.
 - (b) The sample mean and sample median are equal.
 - (c) The sample may have outliers.
 - (d) The population has a symmetric distribution.
3. What is the least possible positive integer-value of n such that $\sqrt{18 \cdot n \cdot 34}$ is an integer?
 - (a) 34
 - (b) 2
 - (c) 17
 - (d) 3
4. Let $M = 4a^2 - 2b^2 + a$. Let j be the value of M when $a = 5$ and $b = 3$, and let k be the value of M when $a = -1$ and $b = 4$. Calculate $j + 2k$.
 - (a) 29
 - (b) 28
 - (c) 27
 - (d) 26
5. What is the sum of all integer solutions to $|n| < |n - 3| < 9$?
 - (a) -9
 - (b) -14
 - (c) 14
 - (d) 9

True / False (10 marks)

Answer True or False

6. True or False: The value of $f(-2)$ for the function $f(x) = ax^6 - bx^4 + x - 1$ is 5.
7. True or False: A cylinder has two circular faces, no vertices, and can roll smoothly on its curved surface.
8. True or False: The correct answer to 'The numbers x and y are inversely proportional. When the sum of x and y is 42, x is twice y . What is the value of y when...' is '40'.
9. True or False: If $3(x - 3) + 10 = 19$, then the value of x is 5.
10. True or False: The solution to the equation $29 = x - 17$ is $x = 46$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the sum of all complex roots of the equation

$$\frac{1}{x-1} + \frac{1}{x-5} + \frac{1}{x-10} + \frac{1}{x-25} = 2,$$

given that there are no repeated roots.

12. Given that Yum-Yum Good candy bars have an average of 210 calories and a standard deviation of 10, analyze the normal distribution to determine what percentage of candy bars exceed 225 calories. Explain your reasoning and calculations.

End of Paper